# Ketan Sharma

+91 6006653315 sharma01ketan@gmail.com **Portfolio** LinkedIn GitHub LeetCode

## Education

**National Institute of Technology Srinagar** **CGPA: 8.6**
*Bachelor of Technology - Information Technology* *2020 – 2024*

## Experience

**Spyne** **May 2025 – Present**
***Software Engineer: Fullstack*** *TypeScript, Go, Node.js, React.js, Electron.js, Aurora, Redis, Kafka, SQS*

- Decoupled the human-in-the-loop review workload from a shared MySQL into a **Go** microservice on **AWS Aurora** (db.r6g.large), serving **5,000+ tenants** at **5K RPM** peaks and making multi-enterprise claims **atomic**.
- Built an image-review queue and **ElectronJS** client on top of it, replacing per-record navigation with pre-assigned batches, cutting handling time from **2.6s to 0.75s** per image and raising 6-hour SLA adherence from **88% to 96%**.
- Extending queue eligibility from one image category to all (**13%** of **300K** daily images), scored on an upstream **GPU** service over **SQS**, with only low-confidence images reaching human review, gated on missed-defect rate.
- Reworked CDN fill from write-time push to on-demand **CloudFront** caching on a **5-minute TTL**, sized against the **28%** actually requesting it, cutting cache footprint **72%** and cost **67%** at **30ms** average render.
- Delivered the agent onboarding and deployment surface on the founding team of a **Conversational AI** product, integrating **VAPI** over **WebRTC/WebSockets** for 5 voice agents with transcription and context-carrying transfer.

**Plane** **August 2024 – May 2025**
***Software Development Engineer: Fullstack - Open Source*** *Next.js, PayloadCMS, Postgres, MobX, S3, HTML, CSS*

- Architected a **PayloadCMS** backend on **Next.js** and **Postgres** with media on **AWS S3** for the marketing site and integrations marketplace, using JSON-driven page components behind custom **RBAC** for partner submitters.
- Migrated the site from **Next.js 14** and **React 18** to **Next.js 15** and **React 19**, moving blogs from SSR to **MDX**-backed **SSG** and lifting Core Web Vitals (LCP, INP, CLS) an average of **25%**.
- Landed **29 merged PRs** and production features in Plane's **50K+ star** open source repository, including authorization fixes and optimistic **MobX** state updates.

**Indian Institute of Technology Jammu** **Feb 2024 – May 2024**
*Machine Learning Intern* *Python, Hugging Face Transformers, BERT, PySyft, TensorFlow*

- Performed model distillation using **5 approaches** on **BERT** and **GPT**-class **transformers**, retaining up to **90%** accuracy, for **PySyft** federated learning on **Android** packet captures.

**Defence Research and Development Organisation** **Dec 2022 – Feb 2023**
*DevOps + Cybersecurity Intern* *Kubernetes, Docker, CTFd, Wireshark, Ansible, Nginx, Linux*

- Led a **team of 3** on a **highly available on-prem Kubernetes cluster on bare-metal Ubuntu servers** for a national CTF, sustaining **300+ concurrent containers** across **400+ participants** from **40+ colleges**.

## Technical Skills

**Languages & Backend**: TypeScript, JavaScript, Go, Python, Java, Rust, Node.js (Nest.js, Express.js), REST APIs, RBAC, Event-Driven Systems (Kafka, SQS), WebSockets, WebRTC
**Cloud, Data & Infrastructure**: AWS (EC2, ECS, Lambda, S3, SQS, CloudFront), Aurora MySQL, PostgreSQL, MongoDB, Redis, Docker, Kubernetes, GitHub Actions, Vercel, Schema Design, Indexing
**Frontend, Testing & AI**: React.js, Next.js, Electron.js, WebAssembly, PayloadCMS, MDX, MobX, Redux/RTK, Tailwind, Jest, Hugging Face Transformers, Agents, LLM, GPU inference, Claude Code, Cursor

## Projects: *Portfolio*

**Dots and Boxes: Rust Game Engine** | *Live Link* | *GitHub* **Rust, WebAssembly, TypeScript, React.js, axum**

- Engineered a **Rust game engine** compiled to **WebAssembly**, with Random, Greedy, **CGT** and **UCT MCTS** opponents plus a **minimax** solver with **Zobrist hasing**, scoped by **PSPACE-completeness** (arXiv 2105.02837).
- Automated CI on **3 GitHub Actions** jobs running **clippy -D warnings**, **65 core tests** and a **wasm** rebuild diff, engines reproducible under a fixed **seed**, built **spec-first** from **15 Linear specs** on **Vercel**.